ITEC 116 – System Integration and Architecture

Group Activity 4: Mid-Project Review & Cross-Module Integration

(Customer Service ↔ Sales & Support)

Group Number: 2

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Scenario Title: Cross-Module Integration — Customer Service ↔ Sales & Support

Focus Modules: Module Customer Service & Module Sales and Customer Support

Sub-Team	Role	Members	Responsibilities
A	**Customer Service Engineers**	Charls Sampaga Kian Lhei Pagkaliwagan Stephanie April Romo	Define support ticket payloads (ticketId, customerId, issueCategory, ticketStatus, priorityLevel).
B	**Sales & CRM Engineers**	Mhar Granado Messiahlyn Mae Mabilangan	Define sales transaction payloads (opportunityId, orderTotal, paymentStatus, accountTier).
C	**Bi-Directional Integration & Mapping Architects**	Curlh Joseph Jordan Golez Laurence Barcarse	Map customer profile fields between CRM and Ticketing databases; define real-time webhook notification triggers.
D	**Mid-Term Systems Audit & QA Lead**	Mike Casaverde Lanz Michael Casino	Audit Modules 1–4 for architectural consistency, evaluate error handling/retry policies, compile the final review presentation.

Section 1: Service & Sales Payload Snippets

Both payloads below are webhook events exchanged between the Sales/CRM system and the Customer

Support ticketing system, keyed on the same customerId so either side can look up the other's record.

1.1 Sales-to-Support Event Payload — Sub-Team B → Sub-Team A

Emitted the moment a deal in Sales_CRM moves to "Closed_Won." The support dashboard listens for this event and automatically creates or updates the corresponding customer profile so agents have full account context before a ticket is ever filed.

```
{
  "eventType": "sale.closed",
  "eventId": "EVT-2026-88213",
  "timestamp": "2026-09-16T10:32:00Z",
  "opportunityId": "OPP-2026-0453",
  "customerId": "CUST-10245",
  "customerName": "BSIT DASH2 Company",
  "contactEmail": "procurement@bsitdash2.com",
```

```
  "orderTotal": 185000.00,
  "paymentStatus": "Paid",
  "accountTier": "VIP_Gold",
  "salesRep": "L. Barcarse",
  "action": "CREATE_OR_UPDATE_SUPPORT_PROFILE"
}
```


1.2 Support-to-Sales Event Payload — Sub-Team A → Sub-Team B

Emitted when a ticket is logged with priority "P1-Critical" or "P2-High." The CRM listens for this event and immediately alerts the assigned account executive, since a high-priority issue on a VIP account can put renewal or upsell conversations at risk.


```
{
  "eventType": "ticket.highPriority",
  "eventId": "EVT-2026-88214",
  "timestamp": "2026-09-16T14:05:00Z",
  "ticketId": "TCK-559012",
  "customerId": "CUST-10245",
  "customerName": "BSIT DASH2 Company",
  "issueCategory": "Service Outage",
  "ticketStatus": "Open",
  "priorityLevel": "P1-Critical",
  "assignedAgent": "M. Mabilangan",
  "accountTier": "VIP_Gold",
  "notifySalesRep": "L. Barcarse",
  "action": "NOTIFY_ACCOUNT_EXECUTIVE"
}
```


Section 2: Customer Profile Mapping Matrix (Task 2)

Source Module & Field	Source Data Type	Target Module & Field	Target Data Type	Integration & Business Logic Rule
Sales_CRM.cust_id	VARCHAR(15)	Cust_Support.client_key	String	Direct key matching for 360-degree profile view.
Sales_CRM.cust_name	VARCHAR(100)	Cust_Support.client_name	String	Direct pass-through; trim whitespace.
Sales_CRM.order_status	INT(1)	Cust_Support.delivery_flag	Boolean	Convert 1 ("Shipped") to true to allow support ticket creation.
Sales_CRM.order_total	DECIMAL(12,2)	Cust_Support.lifetime_value	Number	Add to existing lifetime_value

Source Module & Field	Source Data Type	Target Module & Field	Target Data Type	Integration & Business Logic Rule
				running total on each closed sale.
Sales_CRM.payment_status	VARCHAR(20)	Cust_Support.billing_flag	Boolean	Convert "Paid" to true; "Pending"/"Overdue" to false.
Sales_CRM.assigned_rep	VARCHAR(50)	Cust_Support.escalation_contact	String	Direct pass-through; used to route high-priority ticket notifications.
Cust_Support.sla_tier	VARCHAR(10)	Sales_CRM.account_priority	String	Map "VIP_Gold" to "Priority_1" for account executive alert notifications.
Cust_Support.ticket_priority	VARCHAR(10)	Sales_CRM.alert_flag	Boolean	Convert "P1-Critical" to true to trigger a real-time sales-rep alert.
Cust_Support.ticket_status	VARCHAR(15)	Sales_CRM.open_issue_count	Number	Increment counter when status = "Open"; decrement when "Resolved".

2.1 Real-Time Webhook Notification Triggers

Trigger Event	Firing Condition	Webhook Behavior
Sale Closed	Sales_CRM.deal_stage = "Closed_Won"	Fires sale.closed webhook to Cust_Support within 5 seconds of the database commit; creates or updates the customer's support profile.
High-Priority Ticket Logged	Cust_Support.ticket_priority = "P1-Critical" or "P2-High"	Fires ticket.highPriority webhook to Sales_CRM, notifying the assigned account executive in real time.

Section 3: Mid-Project Systems Audit (Task 3)

3.1 System Consistency Check

Not fully standardized. Auditing the modules built so far surfaces a protocol split:

- Registration/Cashiering module (Activities 1–3): the frontend-to-gateway leg uses REST/JSON, but the gateway-to-legacy-backend leg uses SOAP/XML, since the Cashiering database was modeled as a legacy financial system requiring the stricter SOAP contract.
- Customer Service / Sales & Support module (this activity): both directions use REST/JSON webhooks end-to-end, since both CRM and the ticketing platform are modern systems with no legacy constraint.

3.2 Resiliency & Retry Policy

Error-handling rules for when Customer Service fails to receive a webhook notification from the Sales system (or vice versa):

- Retry schedule: 3 automatic retry attempts using exponential backoff — 5 seconds, 15 seconds, then 45 seconds after the initial failed delivery.
- Timeout: each delivery attempt is aborted and counted as failed if no 2xx acknowledgment is received within 10 seconds.
- Dead-letter queue: if all retries are exhausted, the event is moved to a dead-letter queue instead of being silently dropped, and an alert is sent to the on-call integration engineer.
- Idempotency key: every webhook carries a unique eventId (see Section 1 payloads); the receiving system checks this key before processing so a retried delivery never creates a duplicate ticket or duplicate customer profile update.
- Signature verification: each webhook is signed with an HMAC secret so the receiving system can reject spoofed or corrupted payloads before they are processed.

3.3 Security Protocols Audit

Reviewing the security layer across all four modules built so far:

- Transport security: every module — Registration Portal, Cashiering, and now Sales/Support — communicates exclusively over HTTPS, so this layer is consistent project-wide.
- User-facing authentication: the Registration Portal uses Google OAuth 2.0 (Activity 1–2) so only verified university accounts can submit requests; the same pattern should extend to any human-facing login in the Sales/Support module.
- System-to-system authentication: the new Sales ⇌ Support webhooks add

HMAC request signing (Section 3.2) — a control the earlier SOAP leg does not have today, since it currently relies only on the AuthToken header shown in Activity 2's XML payload.

- Audit recommendation: upgrade the SOAP AuthToken exchange to a signed or OAuth-backed token so all inter-module traffic — not just the new webhooks — shares one verifiable authentication standard.

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Evaluation Rubric (50 Points Total)

- **Bi-Directional Payload Accuracy (15 pts):** Webhook payloads accurately reflect real-time events between Sales and Customer Support.
- **Cross-Module Mapping & Business Logic (15 pts):** Transformation matrix maintains data integrity and supports a 360-degree customer view.
- **Mid-Project Architectural Audit (10 pts):** Clear evaluation of system consistency, resilience, and error-handling policies.
- **Large Group Coordination & Presentation (10 pts):** Clear division of labor among all 8–9 members with a well-structured presentation.

Names of Teams:

A. 1. Charls Sampaga
2. Kian Lhei Pagkaliwagan
3. _____

B. 1. Mhar Granado
2. Messiahlyn Mae Mabilangan
3. _____

C. 1. Laurence Barcarse
2. Curlh Joseph Jordan Golez
3. Stephanie April Romo
4. _____

D. 1. Mike Casaverde
2. Lanz Michael Casino
3. _____